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Assignment Q/A

Q1) what is python and why is it called an interpreted language?

Ans) Python is high-level, interpreted, general purpose programming language known for its easy to read syntex.
It is called an interpreted language because it execute code directly line-by-line via an interpreter rather than requiring a pre-compiled binary.

Q2) what are the key features of python that make it popular for beginners and professionals?

Ans) key features are :-
(i) Simple and Readable Syntax.
(ii) Extensive libraries and frameworks
(iii) Interpreted language
(iv) Versatile and portable
(v) Extensible and integrated
(vi) Free and open source.

Q3) what is the difference between Python 2 and Python 3?

Ans)

Python 2	Python 3
• Python 2 reached its end of life on Jan 1, 2020 it is no longer update or patched for security	• Python 3 is the current, actively supported standard.

Difference	Python 2	Python 3
• Print Syntex	print "Hello"	print("Hello")
• Integer division	3/2 results is 1	3/2 result is 1.5
• string storage	ASCII by default	Unicode by defaut
• Range function	range () to return ~~alist~~ a list and xrange () to return an iterator	range () behave like the old xrange() and xrange() has been removed entirly
• Input handling	raw_input() for string input() to evalute the input	replaced both with single input () function always return str
• Exception handling	used commas and notations	required as key-word for catching exception and use parantheses for the rais statement

Q4) What are python's applications in real-world Projects?

Ans) Major companies use it for recommendation engines (Spotify), data analysis (Instagram), and automation. key application include

building AI models (TensorFlow), web backends ~~Dango~~ (Django/Flask), and data scraping (Beautiful Soup.)

Q5) What is PEP 8 and why is it important in Python programming?

Ans) PEP 8, also known as Python Enhancement Proposal 8, is the official style guide for written for Python code. Created in 2001 by Guido van Rossum, Barry Warsow and Nick Coghlan.
The primary philosophy behind PEP 8 is that "code is read much more often than it is written"

(i) Enhanced Readability
(ii) Improved collaboration
(iii) Easier Maintenance
(iv) Professionalism
(v) Error prevention

Q6) Who developed Python and in which year was it released?

Ans) Python was developed by Guido van Rossum was first released in 1991. The first public version, Python 0.9.0 was released on february 20, 1991.

Q7) What do you mean by "dynamically typed" in Python?

Ans) In Python, dynamic typing means that

that the data type of a variable is determind at runtime (when the program is actually running) rather than at compile time

Q8) what is the difference between a compiler and an interpreter, and which does python use?

Ans)

Feature	Compiler	Python
Execution	Translates entire code at once	Translate line-by-line.
Speed	Faster execution	Slower execution
output	Generates an executable file	No intermediate output file
Errors	Reports all error after scanning	Stops at the first error found
Memory	use more memory	uses less memory
Examples	C, C++, Rust	Python, JS, Ruby

Python is considered an interpreted language because it use an interpreter to execute code line by line.